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Research paper

Modulating the degree of fucosylation of fucosylated chondroitin sulfate enhances heparin cofactor II-dependent thrombin inhibition

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ABSTRACT

Fucosylated chondroitin sulfate (FCS), an unusual glycosaminoglycan with fucose side chains, is a promising anticoagulant agent. To assess the effect of its structure on anticoagulant activity, its derivatives with various degrees of fucosylation (DF), molecular weights (Mw) and sulfation patterns were prepared and characterized. Biological tests showed that their APTT (activated partial thromboplastin time) prolonging activity and intrinsic factor Xase complex (factor IXa-VIIIa-Ca^{2+}-PL complex) inhibitory activity were both reduced in FCS derivatives with lower Mw and DF. However, FCSs with DF at least 16% resulted in greater heparin cofactor II (HCII)-dependent thrombin inhibitory activity in response to decreasing DF, and these activities did not depend on Mw (Mw > 5.2 kDa). Solution competition binding assay further suggested that modulating the DF of FCS derivatives might enhance inhibition of thrombin by activating HCII. These findings imply that FCS derivatives with suitable chain length and DF value may be novel anticoagulants by activating HCII.

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1. Introduction

Heparin cofactor II (HCII) is a plasma serpin that can inactivate thrombin (factor IIa, FIIa), the final protease in the blood coagulation pathway, by forming a bimolecular complex to slow down the coagulation process [1]. This serine protease is involved in the regulation of blood coagulation, atherogenesis and neointima formation. Furthermore, it may help to down-regulate the *in vivo* pathological process of atherosclerosis and thrombosis after vascular injury [2,3]. Unlike antithrombin, HCII-deficient mice develop normally, and no spontaneous thrombosis or other morphological abnormalities have been observed [4,5]. But such mice are more prone to vascular occlusion after arterial injury, suggesting that HCII may have a more important role in response to vascular injury [6].

Some glycosaminoglycans (GAGs), such as dermatan sulfate (DS), heparin and chondroitin sulfate, can accelerate the HCII-thrombin reaction and potentiate the activity of HCII by 1000 times [7,8]. Binding of the GAG to HCII induces its allosteric activation, which increases the accessibility of both the reactive site and an N-terminal acidic domain of the inhibitor to thrombin [9]. Subsequently, DS has been developed as a clinically effective antithrombotic agent owing to its ability to enhance the *in vivo* anticoagulant activity of HCII [10,11].

Fucosylated chondroitin sulfate (FCS) is an unusual GAG that can be extracted from widely traded sea cucumbers (Echinodermata, Holothuroidea). It also exhibits strong thrombin inhibitory activity by heparin cofactor II [7,12]. FCS possesses a chondroitin sulfate-like backbone, but is markedly different from the typical mammalian GAGs because of its unique sulfated fucose side chains (Fig. 1) [12,13]. FCS is a potent anticoagulant through multiple mechanisms. Some mechanisms are well documented such as factor Xa and thrombin inhibition by antithrombin (AT) (AT-dependent anti-FXa and anti-thrombin), inhibition of thrombin by heparin cofactor II (HCII), and inhibition of the intrinsic factor Xase complex (FXase) [14–18]. Previous investigations indicated that the

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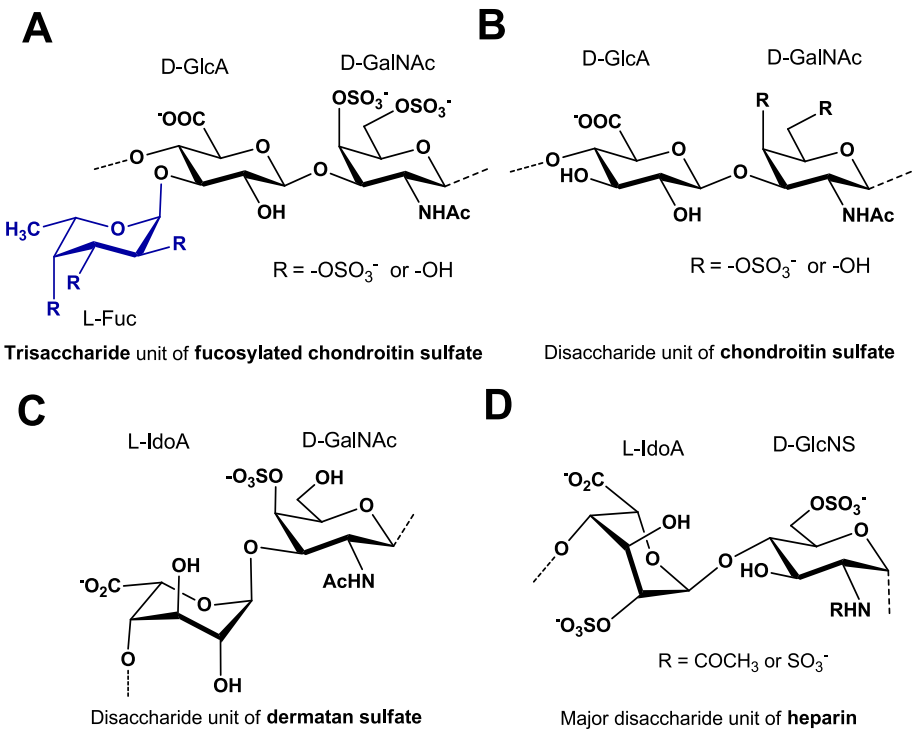


Fig. 1. Major repeating units of several glycosaminoglycans. Regular trisaccharide unit of fucosylated chondroitin sulfate (FCS) from sea cucumber (A) and major disaccharide unit of chondroitin sulfate (CS) (B), dermatan sulfate (DS) (C) and heparin (D).

fucose side chains or sulfate groups are required for the HCII-dependent thrombin inhibitory activity of FCS, since their removal by mild acid hydrolysis significantly weakened this activity [19,20]. Moreover, in the presence of HCII, the thrombin inhibitory activity of fucosylated chondroitin sulfate was about 100-fold higher than that of a chondroitin sulfate [7]. Since the structural backbone units of FCS are the same as those of chondroitin sulfates, the evidence implies that the fucose side chains of FCS are required for its potent thrombin inhibitory activity by HCII [20].

Previously, we studied the relationship between the structure and anticoagulant activity of the GAGs derived from sea cucumbers. We explored the structural modification of native FCS to improve its target selectivity and found that its AT-dependent anti-FXa and anti-thrombin activities could be eliminated by depolymerization, while the HCII-dependent anti-thrombin and the anti-FXase activities remained [12]. Our tests also found that when the molecular weight (Mw) of depolymerized FCS products was above 5 kDa, their HCII-mediated thrombin inhibitory activity increased significantly with the decrease of Mw [12]. The result was curiously inconsistent with other literature reports [21]. Apart from Mw, other structural differences between the FCS derivatives could also be factors contributing to the incongruent findings. Additionally, although research showed that the fucose side chains of FCS contribute to its HCII-dependent anticoagulant activity [19,20], the effect of its degree of fucosylation (DF) on this activity is still unclear. Therefore, its structure-activity relationship, especially the HCII-dependent inhibition activity of thrombin, remains to be clarified.

Factor IXa (FIXa), a serine protease, and factor VIIIa (FVIIIa), a protein cofactor, form a Ca^{2+}- and phospholipid surface-dependent complex referred to as the intrinsic factor Xase complex (FXase), which efficiently converts zymogen factor X to Xa [22,23]. The intrinsic FXase is the last rate-limiting step of the enzyme cascade in the intrinsic coagulation pathway. Consequently, it is becomingly recognized as a prime target for the development of safer anticoagulants [14,18]. Importantly, FCS also has potent intrinsic factor Xase complex inhibitory (anti-FXase) activity. And the depolymerized FCS has weak AT-dependent anti-FXa and anti-thrombin activities, and remains strong anti-FXase activity [12]. Its anticoagulant mechanisms are thus significantly different from those of heparin-like drugs, which have anti-FXa and/or anti-thrombin activities [11]. Hence the depolymerized FCS is becoming recognized as a prime candidate for safer anticoagulants with potential preventive and therapeutic applications [12]. Our previous study further suggested that a minimum of 6–8 trisaccharide units, free carboxyl groups and fucosylation of glucuronic acid (GlcA) residues may be required for potent anti-FXase activity of FCS [12]. However, there has been no detailed investigation about the effects of the degree of fucosylation of FCS on its activity to inhibit thrombin by HCII and the intrinsic factor Xase complex activity.

In this work, the relationships between structure and anticoagulant activity of FCS derivatives were studied to further search for FCS derivatives with selective anticoagulation activity. Specifically, we investigated the effects of degree of polymerization, degree of fucosylation and sulfate substitution types of the fucose side chains on their anticoagulant activities. These APTT (activated partial thromboplastin time) prolonging activities by using normal human plasma were carefully tested to illuminate these anticoagulant activities of FCS derivatives. Furthermore, their HCII mediated thrombin inhibitory activity and intrinsic factor Xase complex inhibitory activity were evaluated to clarify effects of these derivatives on their anticoagulant targets. This study contributes to understanding the structure-activity relationship of the sea cucumber GAG for anticoagulation and the findings are valuable for the development of new anticoagulant agents.

2. Results and discussion

2.1. Preparation and chemical structural characterization of FCS-derived compounds

Previous studies indicated that the glycosidic linkage between branched fucose (Fuc) and glucuronic acid (GlcA) was more susceptible to acid hydrolysis than the linkages in the backbone of FCS [19,20]. Therefore, by modulating the conditions of mild acid hydrolysis based on our previous method (Scheme 1, Table S1) [12,20], seven partially fucosylated products (compounds **2**–**8**) were derived from **1a**, a FCS from the sea cucumber *Thelenota ananas*, with a yield of 40–75%. Further, three partially fucosylated products (compounds **9**–**11**) were obtained from **1b**, a FCS from the sea cucumber *Holothuria fuscopunctata* with a yield of 40–75%.

The chemical structures of **2**–**11** were characterized based on their monosaccharide compositions, functional groups, IR and NMR analyses (see Supporting Information, Table S1, Fig. S1 & S2). That process confirmed that they shared the structural features of backbone chondroitin sulfates, although they differed in the degrees of fucosylation and sulfate content. A single symmetrical peak was observed in the gel filtration chromatography profiles of each product (Fig. S1A & B), indicating the homogeneity of the FCS-derived compounds. The average molecular weight of **1a**, **1b** and their partially fucosylated products ranged from 5.6 kDa to 65.8 kDa, and each one has a narrow distribution of molecular weight (Table S1). According to their ^{1}H NMR spectra in Fig. S3, the integration ratios of signals at ~2.01 ppm and ~1.3 ppm were assigned to protons in the acetyl (3H) in N-acetylgalactosamine (GalNAc) and methyl groups in fucose (3H), respectively [12,24]. The degrees of fucosylation (DF) of the FCS-derived compounds varied enormously, from 2% to 93% (Table 1). The molecular weights of compounds **2**–**11** gradually decreased from 56.0 kDa to 5.6 kDa, suggesting that the FCS backbone might hydrolyse slightly as the defucosylation level increased from 7% to ~98% (DF value decreased).

In contrast to the ^{1}H NMR spectrum of native fucosylated glycosaminoglycan, two signals of the partially defucosylated products **2**–**11** at 3.39 and 3.59 ppm, emerged and increased markedly in response to decreases in DF of the compounds. The signals were assigned to H-2 and H-3 of non-fucosylated glucuronic acid residues. As expected, signals at 5.0–5.7 ppm (assigned to the anomeric protons of the fucose residues) gradually decreased after a mild acid treatment. Additionally, the structures of compounds **2**–**11** were further analyzed based on their ^{13}C NMR spectra (Fig. S4).

The strength of the methyl carbon signal of Fuc at 18 ppm declined gradually in compounds with lower DF. In contrast, the methyl carbon signal of GalNAc at 21.3 ppm varied little, indicating that the chondroitin sulfate-like chain incurred no apparent degradation. Three major anomeric carbon signals were observed at 106.40, 103.71 and 99.79 ppm, which could be assigned to GalNAc, GlcA and Fuc, respectively.

A 2D NMR analysis of compound **8,** with 34% DF, was performed to illustrate the chemical structures of the defucosylated products in detail. The chemical shifts of individual residues were assigned according to 1D (^{1}H, ^{13}C) and 2D COSY, TOCSY, ROESY, HSQC, and HMBC experiments (Fig. S5). The comparison with native FCS and chondroitin sulfate standards [12,13,20], indicated that compound **8** comprises a chondroitin sulfate-like backbone and side chains of sulfated fucose units linked at the C-3 position of GlcA. For compounds **2**–**11** the contents of fucose residues varied from one to another.

The 2D NMR analysis of compound **8** was more complicated compared with its original FCS due to the increased diversity of constituents after partial defucosylation. Based on the analysis of its ^{1}H–^{1}H COSY spectrum, the signals of protons could be confidently assigned and the correlated carbon signals could be observed in the HSQC spectrum. Additionally, the cross peak between GalNAc H1 and GlcA C4 resonances was found in its HMBC spectrum. These results suggest that, through partial mild hydrolysis of FCS under the described condition, some 1 → 3 glycosidic linkages between Fuc and GlcA were cleaved selectively, whereas those in the CS-like backbone chain remained mostly intact.

To further assess how molecular weights of the defucosylated products affect their anticoagulant properties, some low molecular weight compounds (**13**–**17**) derived from compound **12** were prepared by free-radical depolymerization [25]. Their molecular weights of 5.2–27.0 kDa were determined by high performance gel permeation chromatography (HPGPC) (Fig. S1 and Table S1). The DF of these compounds was 52–56% and characterized by ^{1}H NMR spectra, which were similar to that of the initial compound **12**. This narrow range of ^{1}H NMR spectra of compounds **13**–**17** indicated that they had similar chemical structures although their molecular weights differed (Fig. S6 and Table S1). This finding indicated that the backbones of these compounds had glycosidic bonds instead of the Fuc side chain, which were cleaved during depolymerization of compound **12**.

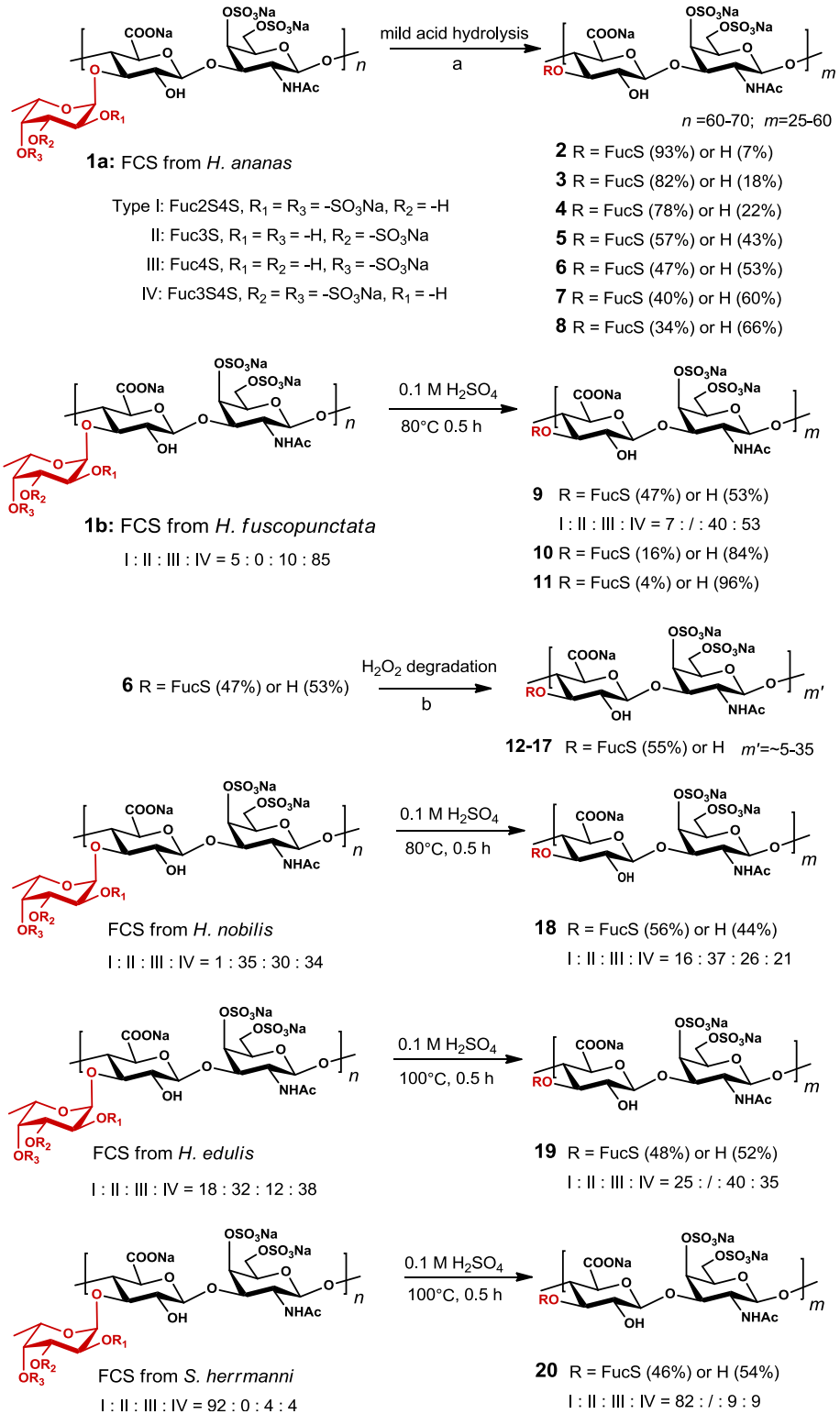
Three partially fucosylated derivatives (compounds **18**–**20**) were prepared in order to investigate the effects of the sulfation types in fucose side chains on anticoagulation activity. Three native FCSs with typically different sulfated fucose types were used to prepare the three derivatives by mild acid hydrolysis [12,26]. Based on their ^{1}H/^{13}C NMR spectra and HPGPC analysis (Fig. S7, Fig. S8 and Fig. S1), the DF of compounds **18**–**20** were similar to each other (46–55%). However, their molecular weights were comparatively dissimilar, ranging between 19.9 and 44.1 kDa. We posit that differences in the sulfate substitution patterns in their fucose branches of native polysaccharides resulted in various hydrolysis of the backbones, though generally acid hydrolysis has little effect on glycosidic bonds in the backbone [20].

All of the products tested were readily soluble in water and thus ready for the anticoagulant analysis in the present study.

2.2. Effects of degrees of fucosylation on APTT prolongation, thrombin inhibition and FXase inhibitory activities

The native FCS and its depolymerized derivatives were found to have potent intrinsic anticoagulant activities [12,14,18,24]. Thus, the effects of DF of FCS derivatives on their anticoagulant activities were evaluated using the activated partial thromboplastin time (APTT). This response is used to determine the ability to inhibit blood clotting through the intrinsic pathways of the coagulation cascade [27]. The results showed that the APTT prolonging activities of compounds **1a** and **2**–**8** from *T. ananas* FCSs, decreased from 125.8 to 14.8 U/mg as their DF decreased from 100% to 34%. Compounds **10** and **11** from *H. fuscopunctata* with lower DF (16% and 2%, respectively) showed weaker activities to prolong APTT of human plasma (6.7 and < 2 U/mg, respectively) (Table 1, Fig. S9A & B). Given the difference of Mw of compounds **1**–**11**, the APTT prolonging activities of **6**–**8** (Mw: 23, 19, 18 kDa; DF: 46, 38, 34%; activity: 38, 19, 15 U/mg) were further compared with **12** and **13** (Mw: 27, 17 kDa; DF: 55, 56%; activity: 56, 42 U/mg) (Tables 1 and 2). These results indicated that fucose branches are required for the anticoagulant effect of FCS on plasma APTT prolongation.

Since inhibition of intrinsic factor Xase complex and thrombin by HCII of FCS derivatives are their two main anticoagulant targets [12,14,18,24], we also tested the above two inhibitory activities of these compounds. When the DFs of FCSs derived from *T. ananas* were at least 78%, the intrinsic FXase inhibitory activity increased



Scheme 1. Synthesis of FCS-derived compounds **2**–**20**.[a]
[a] Reagents and conditions: (a) 100 mM H_2SO_4, 100 °C, 0.5 h, 55%; (b) 30% H_2O_2, $Cu(OAc)_2$, pH 7.5, 35 °C, 2 h, 75–85%.

slightly with lower DF of the compounds (from 248 to 303 U/mg for compounds **1a** and **2**–**4**). In contrast, when DFs were in the range of 30–50% the activity decreased with the reduction of DF values (197–41.9 for compounds **5**–**8**) (Table 1 and Fig. S10A). Compounds **1b** and **9** from *H. fuscopunctata* with respectively 80% and 47% DF, had also strong intrinsic FXase inhibitory activity. In contrast, compounds **10** and **11** from the same source but with DF lower than 30% showed weak FXase inhibition activity (2.21 and < 0.4 U/mg, respectively) (Table 1 and Fig. S10B). Additionally, when the anti-FXase activities of compounds **12**–**13** (Mw: 27, 17 kDa; DF: 55,

Table 1
Effects of degree of fucosylation of FCS on APTT prolongation, HCII-dependent thrombin inhibition and anti-FXase activities.

Compounds	DF (%)	Mw (kDa)	APTT[a]			Anti-thrombin/HCII			Anti-FXase		
			μg/mL	nM	U/mg[b]	IC_{50}		U/mg[b]	IC_{50}		U/mg[b]
						ng/mL	nM		ng/mL	nM	
1a	100	65.8	2.14	32.5	126	402	6.12	57.0	23.7	0.36	248
1b	80	61.1	2.51	41.1	134	295	4.84	136	19.4	0.32	184
2	93	56.0	2.28	40.7	118	341	6.10	67.2	21.9	0.39	268
3	82	49.4	2.91	58.9	92.5	328	6.65	69.9	27.1	0.55	217
4	78	41.0	2.51	61.2	107	247	6.03	92.9	19.4	0.47	303
5	57	30.5	3.98	130	67.7	177	5.82	129	29.8	0.98	197
6	46	23.4	7.07	302	38.1	93.9	4.01	245	47.6	2.03	124
7	38	19.3	14.04	728	19.2	78.0	4.04	294	114	5.89	51.7
8	34	18.3	18.17	993	14.8	80.3	4.39	286	140	7.66	41.9
9	47	39.7	5.08	128	53.0	309	7.80	130	17.4	0.44	206
10	16	15.9	40.18	2527	6.7	104	6.55	386	1624	102	2.21
11	2	5.60	>128	>22857	<2	>10000	>1786	<4	>10000	>1786	<0.4
heparin	–[c]	~18	1.27	70.6	212	108	6.02	212	27.7	1.54	212
LMWH	–[c]	~4.5	6.88	1529	39.1	184	40.8	125	107.5	23.9	54.7
DS	–[c]	41.4	50.8	1227	5.3	37.4	0.902	615	4466	108	1.32
CS-E	–[c]	–	13.12	–	20.5	173	–	133	81.3	–	72.3

[a] The activity of each compound to prolong APTT is expressed as the concentration required to double the APTT (doubling APTT).
[b] The activity of each compound also is expressed as USP units/mg (U/mg) using a parallel standard curve based on the heparin (212 units/mg) from Sigma (USA). The heparin unit of compounds **1b, 9, 10** and **11** was calculated based on the thrombin inhibition IC_{50} of heparin by HCII was 189 ng/mL and the anti-FXase IC_{50} was 16.9 ng/mL.
[c] These compounds don't contain fucose residues.

Table 2
Effects of molecular weight of compounds **12**–**17** on APTT prolongation, HCII dependent thrombin inhibition and anti-FXase activities.

compounds	DF (%)	Mw (kDa)	APTT[a]			Anti-thrombin/HCII			Anti-FXase		
			μg/mL	nM	U/mg[b]	IC_{50}		U/mg[b]	IC_{50}		U/mg[b]
						ng/mL	nM		ng/mL	nM	
12	55	27.0	5.98	222	56.4	257	9.52	156	22.1	0.82	162
13	56	17.0	8.04	473	41.9	218	12.8	185	29.4	1.73	122
14	54	12.2	12.7	1042	26.5	228	18.7	176	44.6	3.65	80.4
15	52	8.80	19.2	2176	17.6	273	31.0	148	83.4	9.47	43.0
16	52	6.90	28.1	4074	12.0	290	42.0	139	191	27.7	18.7
17	51	5.20	46.2	8877	7.3	367	70.6	109	1214	233	3.0
heparin	–	17–19	1.59	88.3	212	190	10.5	212	16.9	0.94	212
DS	–	41.4	64.8	1565	5.2	66.9	1.62	601	5064	122	0.7

[a] The activity of each compound to prolong APTT is expressed by the concentration required to double the APTT (doubling APTT).
[b] The activity of each compound also is expressed as USP units/mg (U/mg) using a parallel standard curve based on the heparin (212 units/mg) from Sigma (USA).

56%; activity: 162, 122 U/mg) were compared with those of compounds **6**–**8** (Mw: 23, 19, 18 kDa; DF: 46, 38, 34%; activities: 124, 52, 42 U/mg) (Tables 1 and 2). This comparison shows that fucosylation was required for the anti-FXase activity of FCS, which decreased with the reduction of their DF when DF is below 78%.

In contrast to the above activities of FCSs, their HCII-mediated thrombin inhibition activity showed a significant increasing trend as their DF decreased. The activity of *T. ananas*-derived compounds **1a** and **2**–**8** increased from 57.0 to 286 U/mg as their DF decreased from 100% to 34% (Table 1 and Fig. 2A). Additionally, among those from *H. fuscopunctata,* compound **10** with 16% DF exhibited the strongest activity (386 U/mg), more potent than that of compounds **9** (47% DF) and **11** (2% DF) (130 and < 4 U/mg, respectively) (Table 1 and Fig. 2B, D). Considering that the difference of their Mw may also affect the results, the thrombin inhibitory activities of compounds **6**–**8** (Mw: 23, 19, 18 kDa; DF: 46, 38, 34%; activity: 245, 294, 286 U/mg) were compared with those of compounds **12** and **13** (Mw: 27, 17 kDa; DF: 55, 56%; activity: 156, 185 U/mg) (Table 2 and Fig. 3A). This comparison further confirmed that their HCII-dependent thrombin inhibitory activities are negatively correlated with the number of sulfated fucose branches.

In summary, the anticoagulant activities of compounds **1a** and **2**–**8** with molecular weight of 18.3–65.8 kDa were converted into the relative activities of heparin (expressed as U/mg in Table 1) and were shown in Fig. 2C. Clearly, in the presence of less sulfated fucose branches (lower DF), their activities of APTT prolongation and intrinsic FXase inhibition diminished. But their capacity to inhibit thrombin by HCII increased, indicating that DF of the compounds has differing effects on their activities.

2.3. Effects of molecular weights on APTT prolongation, thrombin inhibition and FXase inhibitory activities

Further tests were conducted to assess how the molecular weights of FCS affect APTT, the inhibition of intrinsic FXase and thrombin by HCII. Compounds **12**–**17**, which are FCSs with various Mw (27.0–5.2 kDa) and similar DF (~50%) (Table 2), were used in the tests. The concentrations of compounds **12**–**17** required for doubling APTT increased from 6.0 to 46.2 μg/mL as their Mw decreased from 27 kDa (compound **12**) to 5.2 kDa (compound **17**). This result revealed a decline in anticoagulant activity of FCS compounds with lower Mw (Table 2 and Fig. 3A). Similarly, the intrinsic FXase inhibitory activity of FCS derivatives also depends on their molecular weights (Table 2 and Fig. S10C). The IC_{50} values of compounds **12**–**17** increased from 22 to 1214 ng/mL with the reduction of Mw from 27 to 5.2 kDa. At this Mw range their activity was weaker than heparin (IC_{50}, 16.9 ng/mL). Notably, there was a sharp change between compound **16** (6.9 kDa) and **17** (5.2 kDa) (the

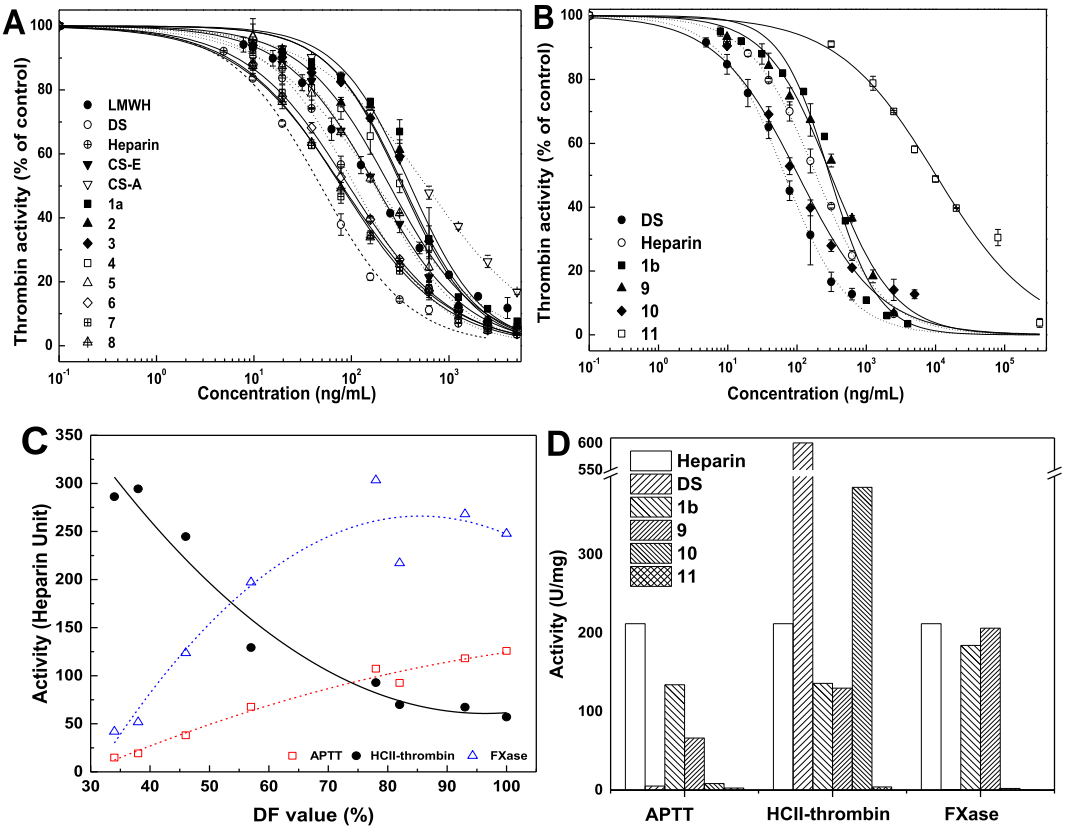


Fig. 2. Effects of native FCS (**1**a and **1**b) and FCS de-fucosylation derivatives (compounds **2**–**8** and **9**–**11**) on thrombin inhibition activity in the presence of HCII (A, B); and the relative activities of each compound on APTT, intrinsic factor Xase activity and thrombin inhibition activity by HCII were converted and expressed as heparin unit (C, D). The results were expressed as mean of duplicate measurements, error bars were added in A and B.

IC_{50} values are 191.4 ng/mL and 1214 ng/mL, respectively).

Interestingly, in contrast to the effects on APTT prolonging and anti-FXase activities, the molecular weight of compounds **12**–**17** did not affect their activity of thrombin inhibition by HCII (Table 2 and Fig. 3A). Instead, their HCII-mediated thrombin inhibitory activity slightly increased with the reduction of Mw on a mass basis. In this case, the IC_{50} values of compounds **12**–**17** increased from 257 to 367 ng/mL, but decreased with the decrease of Mw on a molar basis.

To further illustrate the activity-molecular weight relationship of compounds **12**–**17**, the activities were also converted to heparin unit and examined (Fig. 3C). As the molecular weight decreased from 27 kDa to 5.2 kDa, the ability of compounds **12**–**17** to prolong APTT and inhibit intrinsic FXase decreased. Concomitantly, HCII-dependent thrombin inhibitory activity slightly increased at Mw above 12.2 kDa and then decreased when ≤12.2 kDa. Interestingly, when Mw was as low as 5.2 kDa, compound **17** still displayed strong thrombin inhibitory activity (109 U/mg) by HCII. This activity was much higher than its activity in prolonging APTT (7.35 U/mg) and FXase inhibition (2.95 U/mg).

In our previous study, for FCS depolymerized products with various Mw (not less than about 5 kDa) and approximately 100% DF, the similar effects of molecular weights on the HCII-dependent thrombin inhibitory activity were also observed [12]. Based on the results in the present and previous study, we could conclude that Mw has less effect on HCII-dependent thrombin inhibitory activity for FCS compounds at Mw of above approx. 5 kDa. In contrast, APTT prolonging activity and intrinsic FXase inhibitory activity significantly decreased with the reduction of Mw.

Above all, HCII-dependent thrombin inhibitory of FCS requires much lower DF and Mw than other anticoagulant activities. The FCS samples with low DF (~50%) and Mw (~5–12.2 kDa) exhibited strong and relatively stable HCII-meditated thrombin inhibitory activity. Additionally, their effects on APTT and FXase were weak and significantly affected by Mw. These findings suggest that modulating the degree of fucosylation of sea cucumber FCS with certain chain length could be a way to selectively enhance the inhibition of thrombin by heparin cofactor II.

2.4. Effects of sulfation patterns on APTT, thrombin inhibition and FXase inhibitory activities

To investigate the effects of sulfation patterns on anticoagulant characteristic of FCSs, compounds **9**, **12**, **18**–**20** with similar DF (46–55%) and various sulfated fucose side chains were prepared from native FCS, and their biological activities were tested. APTT assays showed that the activity of compound **18** was about 1.7 times higher than that of **20** (69.5 U/mg for **18** *versus* 39.89 U/mg for **20**) (Fig. S9D). Likewise, a similar effect of sulfation patterns on anti-FXase activity was observed (Fig. S10D). Our pervious result indicated that native compounds (DF about 100%) with higher proportions of Fuc2S4S in the fucose side chain displayed relatively stronger APTT prolonging activity [26]. Both substituents Fuc3S4S and Fuc2S4S in native FCSs might contribute to their strong FXase inhibitory activity, as shown in our previous studies on FCSs from different species [12]. Since Mw of **18** was about 2.2 times higher than that of **20**, it was thus difficult to distinguish whether their sulfation patterns influence the anticoagulant activities.

Among the five samples, the HCII-mediated thrombin inhibitory activity of compound **20** showed the strongest activity, which was

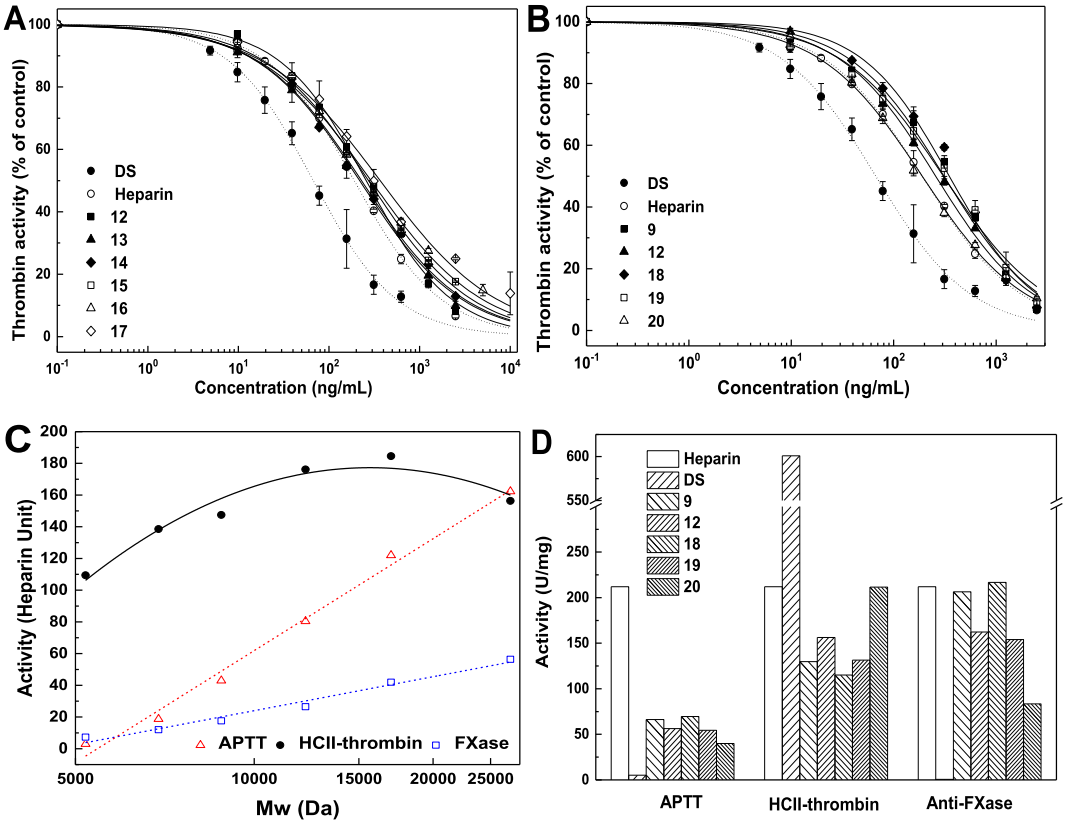


Fig. 3. Effects of Mw, sulfation types in fucose branches of FCS de-fucosylation derivatives (compounds **12**–**17** and **9**, **12**, **18**–**20**) on thrombin inhibition activity in the presence of HCII (A, B); and the relative activities of each compound on APTT, intrinsic factor Xase activity and thrombin inhibition activity by HCII were converted and expressed as heparin unit (C, D). The results were expressed as mean of duplicate measurements, error bars were added in A and B.

about twice higher than **18** (211 U/mg for **20** *versus* 115 U/mg for **18**) (Fig. 3B, D and Table 3). The HCII-mediated thrombin inhibitory activity of compounds **18**–**20** increased as the proportion of Fuc2S4S increased from 16% to 82%. This result implies that FCS with higher sulfate substitution of Fuc2S4S exhibit stronger thrombin inhibitory activity by HCII than those compounds with lower Fuc2S4S branches. Since the above data indicated (Tables 1 and 2) that FCS-derived compounds with lower Mw might exhibit stronger thrombin inhibitory activity by HCII, distinguishing the influence of sulfated fucose types is difficult — the current technology could not eliminate the interference of Mw. Hence, the effect of sulfation pattern on the HCII-dependent thrombin inhibitory activity remains to be elucidated.

2.5. Competitive binding of HCII between the immobilized FCS and soluble FCS derivatives

Interaction of HCII with compound **1a** (Fig. 4) showed that native FCS could bind to HCII with high affinity. The binding model of compound **1a** and HCII fitted well to a 1:1 binding model, and its binding kinetic constants k_{on}, k_{off} and affinity K_D were $7.364 \times 10^3\ M^{-1}s^{-1}$, $3.008 \times 10^{-4}\ s^{-1}$, 4.084×10^{-8} M, respectively.

Table 3
Effects of sulfation patterns in fucose residues of FCSs on APTT prolongation, HCII dependent thrombin inhibition and anti-FXase activities.

Compounds	DF (%)	Mw (kDa)	Fuc Type[a,b] (I: II: III: IV)	APTT[c]			Anti-thrombin/HCII			Anti-FXase		
				μg/mL	nM	U/mg[d]	IC_{50}		U/mg[d]	IC_{50}		U/mg[d]
							ng/mL	nM		ng/mL	nM	
18	56	44.1	16: 37: 26: 21	4.85	110	69.5	349	7.91	115	16.5	0.37	217
9	47	39.7	7:/[e]: 40: 53	5.08	128	66.4	310	7.80	130	17.4	0.44	206
19	48	32.6	25:/[e]: 40: 35	6.20	190	54.4	306	9.38	131	23.3	0.71	154
12	55	27.0	50: 30: 20:/[e]	5.98	222	56.4	257	9.52	156	22.1	0.82	162
20	46	19.9	82:/[e]: 9: 9	8.45	424	39.9	190	9.54	212	42.9	2.16	83.4
Heparin	–[g]	18.0	–[g]	1.59	88.3	212	190	10.5	212	16.9	0.94	212
DS	–[g]	41.4	–[g]	64.78	1565	5.2	66.9	1.62	601	5064	122	0.7

[a] I, Fuc2S4S; II, Fuc3S; III, Fuc4S; IV, Fuc3S4S.
[b] The ratios of distinctive sulfated fucose residues were determined by integration of their anomeric protons of ^{1}H NMR spectra.
[c] The activity of each compound to prolong APTT is expressed by the concentration required to double the APTT (doubling APTT).
[d] The activity of each compound also is expressed as USP units/mg (U/mg) using a parallel standard curve based on the heparin (212 units/mg) from Sigma (USA).
[e] Less than 1%.
[g] These compounds don't contain fucose residues.

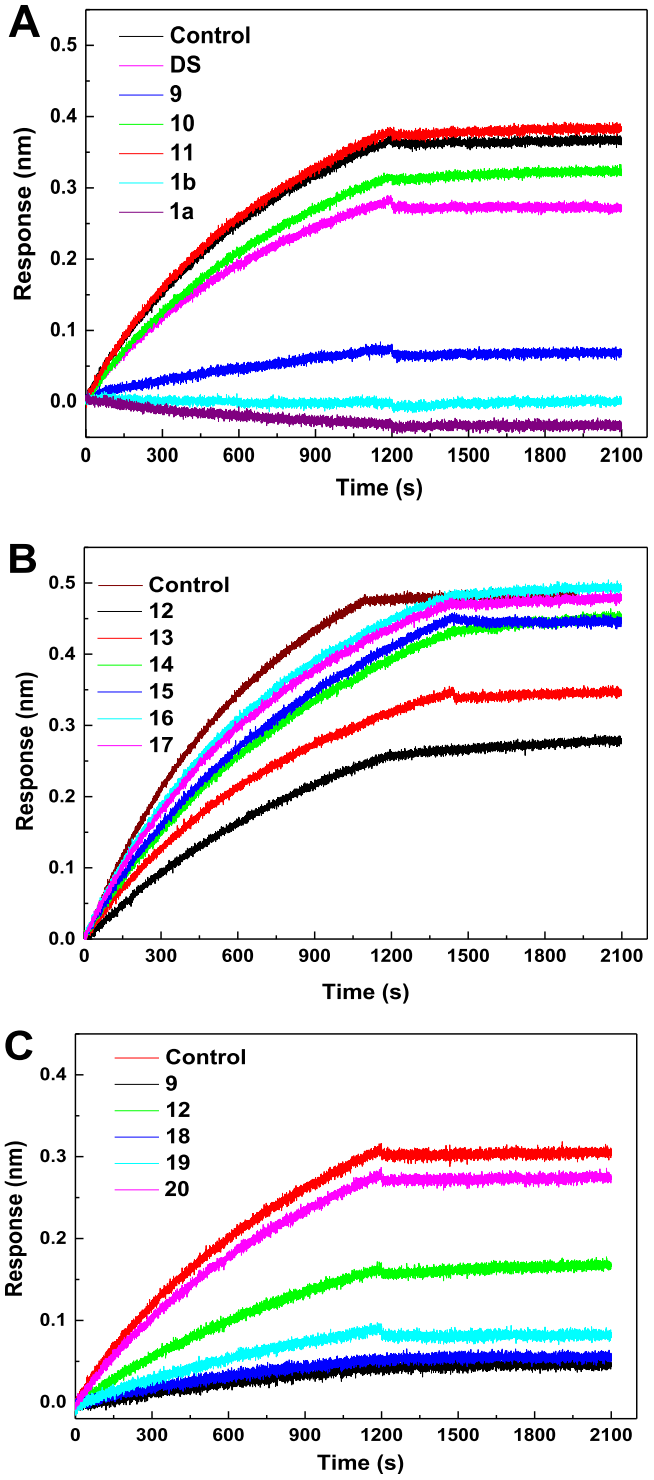


Fig. 4. Competitive inhibition of the binding of HCII and immobilized compound **1a** by soluble FCS-derived compounds (the BLI response signals were expressed as nm).

The data were consistent with our previous observations [28].

To further understand the structure-activity relationship of HCII activation by FCS-derived compounds, their ability to interact with HCII was assessed by a solution-competition assay. In this assay, the activities of soluble GAGs to compete with the immobilized compound **1a** to bind HCII were examined. Compound **1a** at 4 μM were found to completely inhibit the interaction of HCII and compound **1a** (Fig. 4 and Table 4). In comparison, compound **9** with 47% DF and **10** with 16% DF competitively inhibited the interaction by 80.1% and 13.8%, respectively. Moreover, compound **11** with 4% DF did not significantly affect the binding of HCII and immobilized compound **1a**. The results reveal that their competitive binding ability decreased with the reduction of DF value.

Table 4
Competitive inhibition of HCII binding to immobilized compound **1a** by soluble FCS derivatives.

Compounds	Mw (kDa)	DF (%)	Anti-thrombin/HCII[a] (U/mg)	Inhibition (%)
Tris buffer				
1a	65.8	100	57.0	100
1b	61.1	80	136	100
9	39.7	47	130	80.1
10	15.9	16	386	13.8
11	5.60	2	<4	0
12	27.0	55	156	45.6
13	17.0	56	185	27.3
14	12.2	54	176	9.92
15	8.80	52	148	5.25
16	6.90	52	139	0
17	5.20	51	109	0
18	44.1	56	115	83.0
19	32.6	48	131	69.5
20	19.9	46	212	9.64
DS	41.4	–	601	23.1

[a] The concentration of each compound required to inhibit 50% of HCII activity shown as U/mg using a parallel standard curve based on the heparin (212 units/mg) from Sigma (USA).

In contrast, the inhibition rate of the positive compound DS at 4 μM to inhibit HCII to bind immobilized **1a** was only about 23%. DS binds the helix D region of HCII to accelerate the thrombin-HCII reaction [9,29]. Although the compounds **1a** and **9** showed stronger binding ability to HCII than DS, their HCII-dependent inhibition activities of thrombin were lower than that of DS. This finding indicates that compounds **1a** and **9** may bind to other sites (different from the helix D region) with higher affinity. Similar to DS, the compound **10** with DF of 16% exhibited lower ability to bind HCII. On the other hand, it had stronger thrombin inhibitory activity in the presence of HCII than the compounds **1a** and **9** (Table 1). Therefore, the FCS derivatives with low DF values might have a higher selective binding to the D helix domain of HCII. The present study shows that the thrombin inhibitory activity of FCS produced by its binding to HCII may require less than 16% DF in its side chains.

The effects of Mw of the FCS derivatives (with comparable DF) on their HCII binding ability were further tested (Fig. 4 and Table 4). Compound **12** (27.0 kDa) at 4 μM significantly reduced the binding of HCII to immobilized **1a** (the response signal was 0.2610 nm in the presence of **12** *versus* 0.4796 nm of buffer control). This result indicated that **12** could competitively inhibit the binding of HCII and native FCS **1a**, and its inhibition percentage was about 45.6%. At the same concentration, for lower molecular weight compounds **13**–**15**, their competitive inhibition percentages were 27.3%, 9.9% and 5.3%, respectively. By comparison, compounds **16** and **17** with Mw of 6.90 kDa and 5.20 kDa could not interfere the interaction between HCII and immobilized native FCS. The results showed that the ability of FCS derivatives with similar DF to bind to HCII decreased with the decreasing Mw (Fig. 4 and Table 4). This result suggests that, besides the DF, their HCII binding ability also depends on the length of polysaccharide chain.

Notably, the native compound **1b** and its derived compound **9** had similar HCII-dependent thrombin inhibition activities based on a molar concentration (Table 1). Concomitantly, the competitive binding assay showed that the affinity of **9**-HCII interaction was weaker than that of **1b**-HCII. Apart from their strong binding to the

D helix region of HCII, which is the DS binding region leading to allosteric activation of HCII [9,29], FCS derivatives with long chains might also bind to other basic amino acid residues of HCII. This causes an increasing affinity of HCII binding. This might explain that high affinity of HCII binding is not necessarily related to high thrombin inhibitory activity.

Five FCS derivatives **9**, **12**, **18**–**20** with various sulfation patterns and similar DF (47–56%) displayed different abilities of competitive binding to HCII, and their inhibition percentages were 80.1%, 83.0%, 45.6%, 69.5% and 9.6%, respectively (Fig. 4 and Table 4). Given that we could not exclude the influence of their Mw, the effects of sulfation patterns on their abilities of HCII binding remains a subject for further research.

3. Conclusions

In this work, we prepared FCS derivatives with various degree of fucosylation or sulfate substitution in the side chains and various molecular weight. Then, by assessing their APTT prolonging activity, thrombin inhibition and anti-FXase activities, we investigated the structure-activity relationships. Biological tests showed that the APTT prolonging activity and FXase inhibitory activity decreased with the reduction of DF and Mw. However, in contrast to these responses, their thrombin inhibitory activity by HCII increased with the decrease of DF when DF was $\geq$16%. Additionally, the competitive binding assays showed that their HCII-binding ability declined with decreasing Mw and DF of the compounds. This was not necessarily related to their HCII-dependent thrombin inhibitory activity. Nevertheless, these experimental findings suggest that certain chain length and fucosylation level are required for FCS to bind and allosterically activate HCII. Our findings infer that modulating the degree of fucosylation of FCSs could be an effective way to enhance the inhibition of thrombin by heparin cofactor II.

4. Experimental procedures

4.1. Materials

Five fucosylated chondroitin sulfates were extracted from five species of tropical sea cucumbers widely traded in Asia: *Thelenota ananas*, *Holothuria edulis*, *Holothuria nobilis*, *Holothuria fuscopunctata* and *Stichopus herrmanni*. The extraction and purification of the native fucosylated chondroitin sulfates were performed as previously described [24,25]. The yields of the fucosylated chondroitin sulfates were about 0.7–1.25% from the dried body walls of the sea cucumbers.

Heparin (204 IU/mg), dermatan sulfate (DS) and thrombin (123 NIH U/mg) were purchased from Sigma (USA). Low molecular weight heparin (LMWH, 3500–5500 Da, 0.4 mL × 4000 AXaIU) was purchased from Sanofi-Aventis (France). Activated partial thromboplastin time (APTT) reagents were from Teco Medical (Germany). Human heparin cofactor II (HCII) was from Hyphen Biomed (France). Human coagulation factor VIII was from Green Cross China, Inc. (China). The BIOPHEN FVIII: C kit and chromogenic substrate CS-01(38) were purchased from Hyphen Biomed (France). EZ-link amine-PEG_3-biotin and Zeba Spin desalting columns (>7 kDa) were purchased from Thermo Scientific (USA). SA biosensors were purchased from Fortebio (USA). All other chemicals were of reagent grade and obtained commercially.

4.2. General procedures

The ester sulphate content of compounds were determined by turbidimetric method [30]. Optical rotations were determined with a Jasco p-1020 polarimeter. Molecular weights of these compounds were measured by HPGPC using an Agilent technologies 1200 series (Agilent Co., USA) apparatus equipped with a Shodex OH-pak SB-804 HQ column (8 mm × 300 mm) and differential refractive index (RI) detector as described previously [25]. The weight-average molecular mass (Mw) was calculated by GPC software using a standard curve made by dextran standards. For calculating fucosylation degree (DF) of FCS, samples of approx. 5 mg were dissolved in 500 μL of 99.9% D_2O and their ^1H NMR spectra were acquired at room temperature on a Bruker Avance III-600 MHz spectrometer. The degrees of fucosylation (DF) were estimated according to the integral intensities of signals at ~2.01 ppm and ~1.3 ppm, which were assigned to the CH_3CO protons (3H) and methyl groups in fucose (3H), respectively. Compound **8** (30 mg) was dissolved in 1.0 mL of D_2O (99.9%) and lyophilized for three times to replace exchangeable protons with deuterium. The final freeze-dried sample was then dissolved in 500 μL D_2O and its $^1H/^{13}C$, ^1H-^1H COSY, TOCSY, ROESY, ^1H-^{13}C HSQC, HMBC NMR were recorded on a 800 MHz Avance NMR spectrometer.

4.3. Preparation of compounds 2–11 by mild acid hydrolysis

Compounds **2**–**11** were prepared by mild acid hydrolysis procedures of FCS according to modification of the method described previously [20,31,32]. For the obtained FCSs with different degrees of fucosylation (**2** and **3**), about 80 mg of FCS derived from *T. ananas* (**1a**) was dissolved in 16 mL of 0.1 M H_2SO_4 and maintained at 50 °C for different periods of time: 18 min and 24 min for collection of 8 mL. The pH of this solution was adjusted to 7.0 with ice-cold 1.0 M NaOH and then precipitation with 80% ethanol. The hydrolyzed products were obtained as precipitate after centrifugation (4000 r. p. m. for 15 min at room temperature) and these precipitates were dissolved in H_2O. The resulting solution was purified by dialysis (MWCO = 1 kDa) and was then freeze-dried to yield **2** (34.1 mg, 85.2%) and **3** (36.1 mg, 90.2%). Briefly, to obtain compounds **4**–**7,** compound **1a** (200 mg) was dissolved in 40 mL of 0.1 M H_2SO_4 and reacted at 80 °C. The reaction solution (8 mL) was removed after 1, 2, 3, 4 h to yield compounds **4**–**7** with yields of 50–67%. In addition, 80 mg of compound **1a** was hydrolyzed in 16 mL of 0.1 M H_2SO_4 at 100 °C for 0.5 h to prepare compound **8** (35.1 mg, 43.9%). The subsequent purification procedures were similar to the above steps described for **2** and **3**.

Compounds **9**–**11** were obtained by mild acid hydrolysis of FCS derived from *H. fuscopunctata* (**1b**) in 0.1 M H_2SO_4 solution at 80 °C for 0.5 h, 100 °C for 0.5 h and 2 h, respectively.

4.4. General procedure for compounds 12–17

Compound **1a** (4 g) was dissolved in 800 mL 0.1 M H_2SO_4 and maintained at 80 °C for about 2.5 h according to the DF values of compounds **2**–**8**. About 1 g of the resulted product **12** with about degree of fucosylation of 50% was then subjected to further free-radical depolymerizition as previously described [25]. Briefly, compound **12** (1 g) was dissolved in 36 mL H_2O containing 0.5 M NaCl, 0.5 M sodium acetate and 4 mM copper (II) acetate and maintained at 35 °C. 4.0 mL of hydrogen peroxide (10%) was divided into 5 equal parts and was added to the reaction solution at 0, 20, 60, 100, 140 and 180 min, then 7 mL of the solution was withdrawn before the addition of hydrogen peroxide corresponding to compounds **12**–**17** with yield of 50–70%.

4.5. General procedure for compounds 18, 19, 20

Another three native FCSs with different sulfated fucose branches, extracted from the body walls of three sea cucumbers: *H. edulis*, *H. nobilis*, *S. herrmanni* were subjected to mild acid

hydrolysis. *H. nobilis* FCS (150 mg) was dissolved in 30 mL of 0.1 M H_2SO_4 and the solution of was maintained at 80 °C for 30 min under stirring to yield compound **18** (86.63 mg, 57.7%), and *H. edulis* FCS (80 mg) and *S. herrmanni* FCS (90 mg) were kept at 100 °C to yield compounds **19** (44.41 mg, 56%) and **20** (54.5 mg, 60.5%). The subsequent purification procedures were similar to the above steps described.

4.6. Anticoagulant activity assays

APTT, PT, and TT were determined with a coagulometer (TECO MC-4000, Germany) using APTT, PT and TT reagents and standard human plasma as previously described [12,28]. The activity of intrinsic FXase inhibition was determined using the previously described method [12,33] with the reagents in the BIOPHEN FVIII: C kit. The thrombin inhibitory activity in the presence of HCII was detected by the thrombin chromogenic substrate CS-01(38) using a Bio-Tek Microplate Reader [12,33].

4.7. Solution competition binding assay

The native FCS (compound **1a**) from *T. ananas* was biotinylated by using amine-PEG_3-biotin as previously described [34,35]. Further, the reaction mixture was desalted with the Zeba Spin desalting columns. Then the biotinylated **1a** was immobilized onto the surface of SA biosensors according to our previous method [28]. 250 nM HCII was allowed to interact with immobilized **1a**.

All interaction experiments were conducted at 30 °C in PBSB (0.05 M sodium phosphate at pH 7.2, 0.15 M NaCl, 0.1% BSA) using an Octet Red 96 instrument (Fortebio, USA). Final volume for all the solutions was 200 μL. Assays were performed in black solid 96-well flat bottom plates with agitation set to 1000 r.p.m. A 600s biosensor washing step was applied prior to the analysis of the association of the ligand on the biosensor to the analyte in solution for 1200 s. Finally, the dissociation was followed for 900 s. Dissociation buffer solutions were used only once to avoid potential contamination. After dissociation, the sensor surface was regenerated in 4 M NaCl. Correction of any systematic baseline drift was done by subtracting the shift recorded for a sensor loaded with ligand but incubated with no analyte. After each run, dissociation and regeneration were performed as described above. To assess the relative ability of soluble compounds to compete with the immobilized native polysaccharide FCS (compound **1a)** for binding to HCII, a competition binding assay was performed to determine their response values. HCII (250 nM) was preincubated with 4 μM each compound prior to interact with immobilized **1a**. For BLI kinetic assays, data were analyzed using the Octet software version 7.0 and the binding curves were globally fitted using a 1:1 model [36].

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ejmech.2018.05.024.

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